Overview of Chapter 11

Geometry in Three Space and Vectors

1. Dot Product (Scalar Product)

(1) Definition

$$\boxed{\mathbf{u}\cdot\mathbf{v}=\langle u_1,u_2,u_3\rangle\cdot\langle v_1,v_2,v_3\rangle=u_1v_1+u_2v_2+u_3v_3}$$

(2) Properties

If $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are vectors, and c is a scalar, then

1. $\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u}$
2. $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w}$
3. $c(\mathbf{u}\cdot\mathbf{v})=(c\mathbf{u})\cdot\mathbf{v}$
4. $\mathbf{0}\cdot\mathbf{u}=0$
5. $\mathbf{u}\cdot\mathbf{u}=\|\mathbf{u}\|^2$

(3) Geometric Meaning

If θ is the smallest nonnegative angle between the nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u}\cdot\mathbf{v}=\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$$

(4) Applications of Dot Product

1. Computing Lengths and Angles $\cos\theta=\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$, $\theta\in[0,\pi]$.

2. Determine whether two vectors are perpendicular

$$\mathbf{u}\perp\mathbf{v}\Leftrightarrow\mathbf{u}\cdot\mathbf{v}=0$$

3. Direction Angles and Cosines

If $\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}$, then

$$\cos\alpha=\frac{\mathbf{a}\cdot\mathbf{i}}{\|\mathbf{a}\|\|\mathbf{i}\|}=\frac{a_1}{\|\mathbf{a}\|},\cos\beta=\frac{\mathbf{a}\cdot\mathbf{j}}{\|\mathbf{a}\|\|\mathbf{j}\|}=\frac{a_2}{\|\mathbf{a}\|},\cos\gamma=\frac{\mathbf{a}\cdot\mathbf{k}}{\|\mathbf{a}\|\|\mathbf{k}\|}=\frac{a_3}{\|\mathbf{a}\|}.$$

4. Projection of u on v

Vector projection: $\mathrm{pr}_{\mathbf{v}}\mathbf{u}=\left(\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\right)\mathbf{v}$

Scalar projection: $\|\mathbf{u}\|\cos\theta=\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$

2. Planes in 3-Space

(1) Standard Equations

The equation of the plane through the point $P(x_0, y_0, z_0)$ and perpendicular to $\mathbf{n} = \langle A, B, C \rangle$ (normal vector) is

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0 \Leftrightarrow Ax + By + Cz = D$$

where $D = Ax_0 + By_0 + Cz_0$.

(2) Distance from Point to Plane

The distance L from the point (x_0, y_0, z_0) to the plane $Ax + By + Cz = D$ is given by the formula

$$L = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}.$$

3. Cross Product (Vector Product)

(1) Definition

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

(2) Geometric Meaning

Direction: $\mathbf{u} \times \mathbf{v} \perp \mathbf{u}, \mathbf{u} \times \mathbf{v} \perp \mathbf{v}$

$\mathbf{u}, \mathbf{v},$ and $\mathbf{u} \times \mathbf{v}$ form a right-handed triple

Magnitude $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin\theta$

(3) Applications of Cross Product

1. Determine whether two vectors **u** and **v** are parallel?

$$\mathbf{u} // \mathbf{v} \Leftrightarrow \mathbf{u} \times \mathbf{v} = \mathbf{0}$$

2. Find a vector perpendicular to both **u** and **v**.

3. Find the area of a parallelogram with **u** and **v** as adjacent sides.

$$Area = \|\mathbf{u} \times \mathbf{v}\|$$

4. Find the volume of the parallelepiped determined by the vectors with **a**, **b** and **c**.

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \left\| \begin{matrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{matrix} \right\|$$

(4) Algebraic Properties

Anti-commutative law $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$

4. Lines in 3-Space

Equations

The **parametric equations** of the line through the point $P(x_0, y_0, z_0)$ and parallel to $\mathbf{v} = \langle a, b, c \rangle$ (direction vector) are

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

The **symmetric equations** are

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}, \text{ if } abc \neq 0.$$

4. Vector Valued Functions

(1) Calculus of Vector Valued Functions

(2) Parametric Equations for Curves in 3-Space

Tangent line to a curve

Let $\mathbf{r} = \mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = \langle f(t), g(t), h(t) \rangle$ be a curve in three-space. The tangent line to the curve has direction vector

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k} = \langle f'(t), g'(t), h'(t) \rangle.$$

(3) Curvilinear Motion

Velocity $\mathbf{v}(t) = \mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$,

Acceleration $\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}$.

Arc Length $s = \int_a^t \sqrt{[f'(u)]^2 + [g'(u)]^2 + [h'(u)]^2}\, du = \int_a^t \|\mathbf{r}'(u)\|\, du$.

Speed $\dfrac{ds}{dt} = \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} = \|\mathbf{r}'(t)\|$.

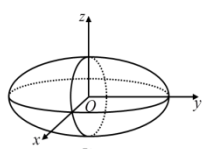
5. Surfaces in 3-Space

(1) Cylinder

If one of the variables x, y or z is missing from the equation, then the equation represents **a cylinder in three-space .**

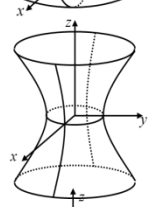
(2) Quadric Surfaces

1. Ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$



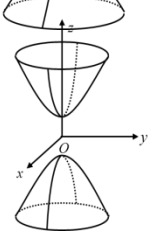
2. Hyperboloid of One Sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

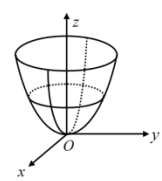


3. Hyperboloid of Two Sheets

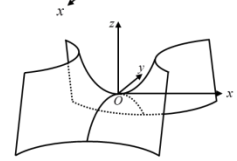
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$



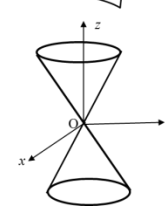
4. Elliptic Paraboloid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = z$



5. Hyperbolic Paraboloid $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = z$



6. Elliptic Cone $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 0$



Overview of Chapter 12

Derivatives for Functions of Two or More Variables

1. Functions of two or more variables

(1) Graphs and Level Curves of $f(x, y)$

(2) Level Surfaces of $F(x, y, z)$

2. Limit and Continuity

Two Path Test

If $f(x, y) \to L_1$ as $(x, y) \to (a, b)$ along a path C_1 and $f(x, y) \to L_2$ as $(x, y) \to (a, b)$ along a path C_2, where $L_1 \neq L_2$, then $\lim\limits_{(x,y)\to(a,b)} f(x, y)$ does not exist.

3. Partial Derivatives

(1) Notations

$$f_x(x\ ,y\ ) = \frac{\partial z}{\partial x} = \frac{\partial}{\partial x} f(x, y), \quad f_y(x\ ,y\ ) = \frac{\partial z}{\partial y} = \frac{\partial}{\partial y} f(x, y)$$

$$f_x(x_0, y_0) = \left.\frac{\partial z}{\partial x}\right|_{(x_0, y_0)}, \qquad f_y(x_0, y_0) = \left.\frac{\partial z}{\partial y}\right|_{(x_0, y_0)}$$

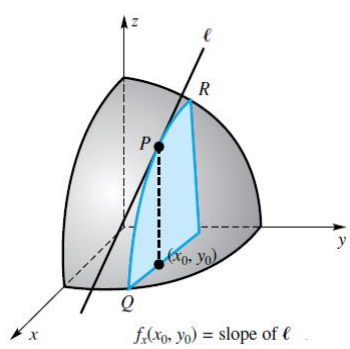
∂ is called the partial derivative sign.

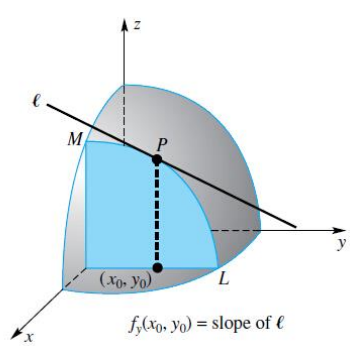
(2) Computation

Rule for Finding Partial Derivatives of $z = f(x, y)$.

1. To find f_x, treat y as a constant and differentiate $f(x, y)$ with respect to x.
2. To find f_y, treat x as a constant and differentiate $f(x, y)$ with respect to y.

(3) Geometric Meaning





(4) Higher Partial Derivatives

$$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} \qquad f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}$$

$$f_{xy} = (f_x)_y = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} \qquad f_{yx} = (f_y)_x = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

4. Directional Derivatives

(1) Definition

Let $P_0(x_0, y_0)$ be a given point . For any unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$,

let $$D_{\mathbf{u}} f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + hu_1, y_0 + hu_2) - f(x_0, y_0)}{h}.$$

This limit, if it exists, is called the ***directional derivative*** of f at $P_0(x_0, y_0)$ in the direction of $\mathbf{u}$.

(2) Computation

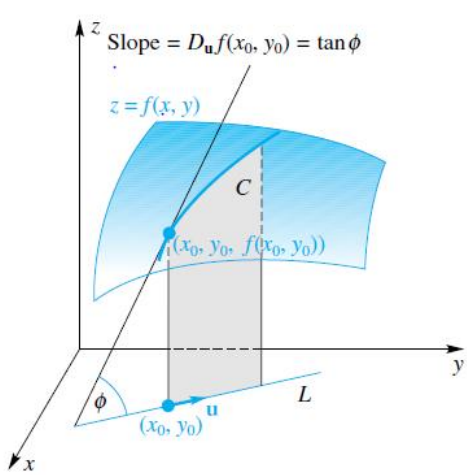
The ***directional derivative*** of f at $P_0(x_0, y_0)$ in the direction of unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u}.$$

(3) Practical Meaning

The directional derivative $D_{\mathbf{u}} f(x_0, y_0)$ measures the rate of change of $f(x, y)$ with respect to distance in the direction of $\mathbf{u}$ at the point $P_0(x_0, y_0)$.

(4) Geometric Meaning



5. Gradient

(1) Definition

The ***gradient*** of $f(x, y)$ at $P_0(x_0, y_0)$ is

$$\nabla f(x_0, y_0) = f_x(x_0, y_0)\mathbf{i} + f_y(x_0, y_0)\mathbf{j}$$

The ***gradient*** of $F(x, y, z)$ at $P_0(x_0, y_0, z_0)$ is

$$\nabla F(x_0, y_0, z_0) = F_x(x_0, y_0, z_0)\mathbf{i} + F_y(x_0, y_0, z_0)\mathbf{j} + F_z(x_0, y_0, z_0)\mathbf{k}$$

(2) Properties of Gradient

If $\nabla f \neq \mathbf{0}$, then function $f(x, y)$ increases most rapidly at $P(x_0, y_0)$ in the direction of the gradient $\nabla f(x_0, y_0)$ with rate of change $\|\nabla f(x_0, y_0)\|$ and decreases most rapidly in the opposite direction $-\nabla f(x_0, y_0)$ with rate of change $-\|\nabla f(x_0, y_0)\|$.

Direction of Gradients

- **Perpendicular to the level curves**
- **Point higher values of the function (the direction of greatest increase of the function)**

6. Differentiability

(1) Definition

The function $z = f(x,y)$ is said to be ***differentiable*** or ***locally linear*** at (x_0, y_0) if the increment $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$.

can be expressed in the form

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$$

where $\varepsilon_1 \to 0$ and $\varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$.

(2) Local Linear Approximation

$$\boxed{f(x,y) \approx f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) = L(x,y).}$$

(3) Total differential

$$\boxed{dz = f_x(x,y)dx + f_y(x,y)dy}$$

(4) Tangent Plane

A normal vector and the equation of the tangent plane to $z = f(x,y)$ *at* $P\left(x_0, y_0, f(x_0,y_0)\right)$ *are*

$$\mathbf{n} = \left\langle f_x(x_0,y_0), f_y(x_0,y_0), -1 \right\rangle$$

$$f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) - (z - f(x_0,y_0)) = 0$$

A normal vector and the equation of the tangent plane to $F(x,y,z) = 0$ *at* $P\left(x_0, y_0, z_0\right)$ *are*

$$\mathbf{n} = \nabla F(x_0,y_0,z_0) = \left\langle F_x(x_0,y_0,z_0), F_y(x_0,y_0,z_0), F_z(x_0,y_0,z_0) \right\rangle$$

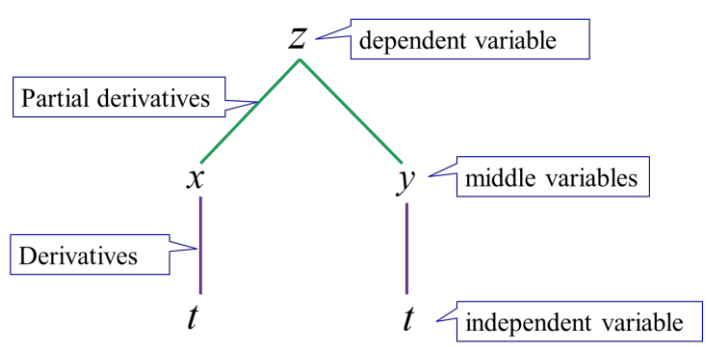
$$F_x(x_0,y_0,z_0)(x-x_0) + F_y(x_0,y_0,z_0)(y-y_0) + F_z(x_0,y_0,z_0)(z-z_0) = 0$$

7. Chain Rules

(1) First Version

If $z = f(x, y)$, where $x = g(t)$, $y = h(t)$, then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

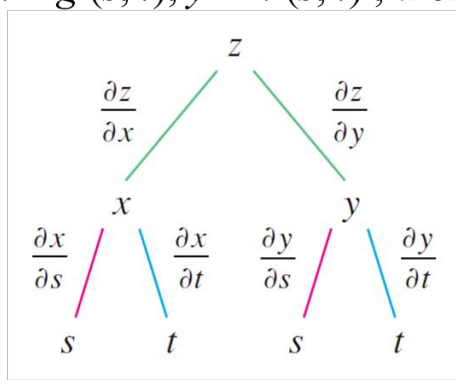


(2) Second Version

If $z = f(x, y)$, where $x = g(s, t)$, $y = h(s, t)$, then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s},$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}.$$



(3) Implicit Differentiation

Case I Suppose that $F(x, y) = 0$ defines an implicit function $y=y(x)$. If $F_y \neq 0$, then

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Case II Suppose that $F(x, y, z) = 0$ defines an implicit function $z=z(x,y)$. If $F_z \neq 0$, then

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

8. Applications of Partial Derivatives

(1) Local Extreme Values

First Partials Test for Local Extreme Values

If f has a local maximum or minimum value at an interior point (a,b) of its domain and if the first partial derivatives exist there, then $f_x(a,b) = 0$ and $f_y(a,b) = 0$,

At such point the tangent plane is horizontal.

Second Partials Test for Local Extreme Values

Suppose that $f(x,y)$ has continuous second partial derivatives in a neighborhood of (a,b) and that $\nabla f(a,b) = \mathbf{0}$. Let

$$A = f_{xx}(a,b),\ B = f_{xy}(a,b) = f_{yx}(a,b),\ C = f_{yy}(a,b)$$

$$D = AC - B^2 = f_{xx}(a,b) f_{yy}(a,b) - (f_{xy}(a,b))^2$$

Then

(i) if $D > 0$ and $A < 0$, $f(a,b)$ is a **local maximum value**;

(ii) if $D > 0$ and $A > 0$, $f(a,b)$ is a **local minimum value**;

(iii) if $D < 0$, $f(a,b)$ is not an extreme value ((a,b) is a **saddle point**);

(iv) if $D = 0$, the test is **inconclusive**.

Procedure for Finding Local Extrema of $z = f(x,y)$

Step 1 Solve the simultaneous equations $\begin{cases} f_x(x,y) = 0 \\ f_y(x,y) = 0 \end{cases}$

to identify all stationary points.

Step 2 Calculuate the second partial derivatives f_{xx}, f_{xy}, f_{yy} and $D(x,y)$ and evaluate the values of f_{xx} and $D(x,y)$ at each stationary point, then list a Table.

Step 3 Use the results in Table and Theorem C to determine whether or not f has local extreme value at each stationary point and evaluate local extreme values of f.

(2) The Method of Lagrange Multiplier

Lagrange's Method

To find extremum values of $f(x,y)$ subject to the constraint $g(x,y) = 0$:

Step 1 Solve the system of equations $\begin{cases} \nabla f(x,y) = \lambda \nabla g(x,y) \\ g(x,y) = 0 \end{cases}$

to identify all critical points.

Step 2 Evaluate f at all critical points (x,y). The largest of these values is the maximum value if exists; the smallest is the minimum value if exists.